

Training Language Models to Follow Instructions with Human Feedback

Summary

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1 Problem Setup

Large language models like GPT-3 can generate fluent text but often fail to follow human intentions and preferences. They may produce harmful content, refuse harmless requests, or simply not do what users want. This paper introduces InstructGPT (the model behind ChatGPT), which uses reinforcement learning from human feedback (RLHF) to align language models with human preferences and make them more helpful and harmless.

2 Thesis

The central thesis is that reinforcement learning from human feedback can effectively align large language models with human preferences, making them more helpful and safer without requiring explicit supervision for every possible task. By training a reward model from human preferences and using it to fine-tune the language model, we can dramatically improve alignment while maintaining performance.

3 Key Concepts

Reinforcement Learning from Human Feedback (RLHF): A three-step process where humans label preferences, a reward model learns from these preferences, and the language model is fine-tuned using reinforcement learning with the reward model as the reward function.

Reward Model: A language model trained to predict which of two responses humans would prefer. It serves as a proxy for human judgment during reinforcement learning training.

PPO (Proximal Policy Optimization): The reinforcement learning algorithm used to fine-tune the language model. It updates the model to maximize the reward while staying close to the original model to avoid catastrophic forgetting.

Instruction Following: The ability of language models to understand and execute user instructions, as opposed to simply continuing text patterns.

4 Methods

The training process consists of three main steps:

1. **Supervised Fine-Tuning:** First, the pre-trained language model is fine-tuned on human-written demonstrations of instruction following. This creates an initial model that can follow instructions.

2. **Reward Model Training:** Human labelers compare different model responses to the same prompt and indicate which they prefer. These preferences are used to train a reward model that predicts human preferences.
3. **RL Fine-Tuning:** The language model is fine-tuned using PPO with the reward model providing the reward signal. The model is encouraged to generate responses that humans would prefer while staying close to the original model.

The process is iterative, with new human preferences collected continuously to improve the reward model and prevent reward hacking.

5 Results

The authors evaluate InstructGPT models on several benchmarks:

- **Human Preference Evaluations:** InstructGPT outputs are preferred to outputs from the original GPT-3 model 85% of the time, even when the original model is 10x larger.
- **Truthfulness:** InstructGPT models show improved truthfulness compared to GPT-3, with reduced fabrication of information.
- **Helpfulness:** The models are rated as more helpful by human evaluators across a wide range of tasks.
- **Safety:** InstructGPT produces fewer harmful responses and is better at refusing inappropriate requests.
- **Performance Trade-offs:** While alignment improves, there are small decreases in performance on some traditional NLP benchmarks, but these are outweighed by the improvements in human preference.

The results show that RLHF can effectively align language models with human preferences across diverse tasks and domains.

6 Significance

This work represents a major breakthrough in AI alignment:

- **Practical Alignment:** Demonstrates a scalable method for aligning large language models with human values and preferences.
- **Foundation for ChatGPT:** The techniques described in this paper are the foundation for ChatGPT and similar conversational AI systems.
- **Industry Standard:** RLHF has become the standard approach for training helpful and harmless language models.
- **Scalability:** The method works for models ranging from 1.3 billion to 175 billion parameters, suggesting it can scale to even larger models.
- **Reduced Supervision:** The approach requires far less explicit supervision than traditional fine-tuning, making it more practical for real-world deployment.

The paper established RLHF as a fundamental technique for AI alignment and paved the way for the current generation of more helpful and safer language models.